Software Engineering Group Project

COMP2002: Project Handbook 2024-2025

Table of Contents

1 Introduction 5
1.1 Overview 5
1.2 BCS Accreditation 6
1.3 Ethics 7
1.4 Non-Disclosure Agreements and Intellectual Property 9
1.5 Workload 10
1.6 Supporting Lectures 11
1.7 Teams 12
1.8 Roles and Responsibilities 13
1.8.1 The Group Project Coordinator 13
1.8.2 Supervisor Roles 13
1.8.3 Team Roles 14
1.9 Expenses 16
1.9.1 Travel 16
1.9.2 Equipment 16
1.10 Background Reading 17
2 The Project 18
2.1 The Bidding Process 18
2.1.1 Project Sign-Up 18
2.1.2 Bidding for a Project 18
2.1.3 Award Process 19
2.2 Project Management 20
2.2.1 “Backlogs” 20
2.2.2 “Sprints” 20
2.2.3 “Stand-Ups” or “Scrums” 21
2.2.4 Supervisor Meetings 21
2.3 Version Control - Git 22
2.4 Code quality 23
2.5 Testing 24
2.6 Open Day 25
2.7 End-Of-Contract 26
2.8 Golden Rules 27
3 Group Reports 28
3.1 Submission Format 28
3.2 General Advice 29
3.2.1 Writing Process 29

3.2.2 Content ... 29
3.2.3 Language & Style ... 29
3.2.4 Structure ... 29
3.2.5 Referencing ... 30
3.3 Legal, Societal, Ethical and Professional Issues (LSEPI) ... 31
3.3.1 Context ... 31
3.3.2 “Big picture” Reflection Checklist ... 31
3.4 The Interim Report ... 33
3.5 The Final Report ... 34
3.5.1 Main Report ... 34
3.5.2 Software Manual ... 35
3.5.3 User Manual ... 35
3.5.4 Demonstration Video ... 36
3.5.5 Abstract ... 36
4 Individual Reports ... 37
4.1 Individual Report - Reflective Statement ... 37
4.2 Peer Assessment ... 38
5 Assessment ... 40
5.1 Summary of Deliverables ... 40
5.2 Group Marks ... 40
5.3 Individual marks ... 40
5.4 Total marks ... 41
5.5 Re-assessment ... 42

Preface

This project handbook is based on the work done in prior years by Peter Blanchfield and other colleagues. It is still evolving. That is, some sections may still be added, typos and grammar corrected, and further clarification may still be needed in places. All thoughts and feedback to improve it will be greatly appreciated.

1 Introduction

1.1 Overview

The software engineering group project module brings students together to work in team on a project. The module and project focus on the entire software development lifecycle, including the software specification (requirements analysis and design), development (implementation, coding, programming), validation and testing, documentation, and the management of the project. That is, this module is not “just” about writing code, but instead about the whole software engineering (SE) process. This will require teams to bring theory that they have learned (or will learn) in other modules, including G51FSE G51PGP, G51PGA, and G52SWM, into practice. Supporting lectures and project supervisor(s) will guide you through the process.

The module consists of the following key elements:

- The **bidding process** involving the submission of your team’s CVs, Expressions of Interest (EoI), a set of formal pitches, and a project award process.
- The **project** itself, applying technical knowledge, established SE methods, agile project management, and interpersonal skills to successfully deliver the project.
- The **demo day** (circumstances permitting), a "trade fair style event" where you will demonstrate your work to both internal and external visitors.
- The **delivery / end of contract**, including the submission of your software, project report, user and software documentation, management documentation, an individual project report, and a peer review.

For many of you, the group project will offer a first working experience on a team based project, helping you to apply theoretical and practical knowledge to deliver a real result. The group project is generally regarded as one of the most important components of your UG degree and encapsulates key elements towards the BCS accreditation of your degree. Many alumni comment to us that this module has had the most value to them as they start their working life.

1.2 BCS Accreditation

The BCS (British Computing Society) is the Chartered Institute for IT, the UK's main professional association for IT. It manages and regulates professional accreditation for IT and computing. The BCS accreditation of a degree shows that it covers the core material and skills needed for professional accreditation. Note that for accreditation the BCS also requires that you do – and pass – a third year project, and that you do your third year at a University of Nottingham campus (UK, Malaysia or China)[1].

The criteria required for BCS accreditation are listed here. Criteria covered in the group projects module include (but are not limited to):

- Knowledge of management techniques to achieve objectives.
- Work as a member of a development team.
- Development of general transferable skills.
- Deploy theory in design, implementation and evaluation of systems.
- Recognise legal, social, ethical & professional issues.
- Knowledge and understanding of commercial and economic issues.
- Knowledge of management techniques to achieve objectives.
- Knowledge and understanding of engineering principles.

[1] Information provided here is paraphrased from the document here.

1.3 Ethics

The University of Nottingham requires all students engaged in work that involves human participants or human data (including biological data) to complete an Ethics Review. **All groups must complete the CS Preliminary Ethics Checklist.** For most projects, completing the preliminary ethics checklist will be sufficient.

Students should **work together with their academic supervisors** to complete the checklist. After completion and agreement from the supervisor, **each group submits the preliminary ethics checklist to the relevant submission system in Moodle** (one per group, submitted by the group administrator) and notify their supervisor of the submission. The supervisor will check and "mark" the submission.

If **your project involves, e.g., human participants, the use of personal data that is not in the public domain, or involves biological materials, your team will also need to complete the full CS Research Ethics Checklist.** After completion and agreement from the supervisor, the team should submit the full ethics form and supporting documentation to the relevant submission system in Moodle. The submission will be checked and approved by the supervisor, or sent to the CS ethics committee for further review. You will be required to revise your ethics application if it does not satisfy the ethics requirements.

Groups must not start work with human participants, personal data that is not in the public domain, or biological materials before the ethics review process has been completed. You are therefore advised to start the ethics process as soon as possible (i.e. immediately after you have been assigned your project or as soon as it becomes apparent that you need ethics clearance).

The forms necessary to complete for the ethics process are available on Moodle and on the school's Ethics Forms on workspace.

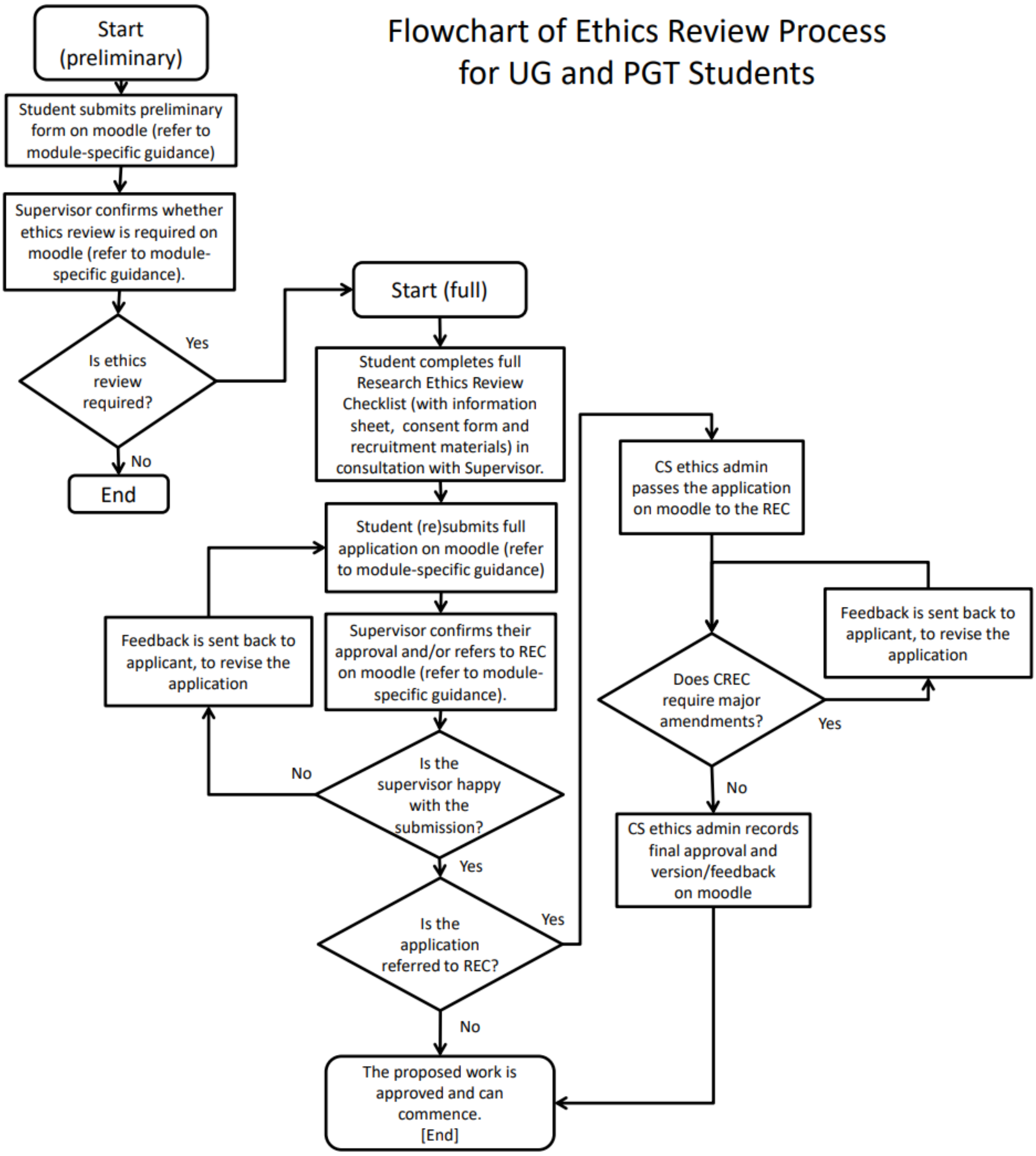

Figure 1: Ethics Checks - Flow chart (original available on the CS workspace)

1.4 Non-Disclosure Agreements and Intellectual Property

Those working on projects with **commercially sensitive** or **personal data** will be covered by non-disclosure agreements with their sponsor. As a result, **no data or information pertaining to your project should be kept in a publicly accessible form unless it is in line with the terms of the NDA**. The template for the non-disclosure agreement is provided on Moodle.

If you should develop IP that the sponsor you are working with should want to exploit, please be aware that the **students are themselves owners of this IP** and will ultimately need to negotiate the transfer of this to the company. That is, if the outcome of your project has commercial potential, then a separate agreement will govern its exploitation. The University and School will work with you if you wish to ensure this is done legally. Note that there is also the Nottingham Ingenuity Lab to help University of Nottingham students and alumni to develop their entrepreneurial ambitions.

Additional information on intellectual property can be found here.

1.5 Workload

COMP2002 is a 20 credit module. This means that each member of the team should spend roughly 200 hours of work on the project, or in a team of 6 people, a total of 1200 hours. That is, one could reasonably expect to work about 100 hours per term on their group project, or on average just over 1 day per week in a 12 week term. Note that the group marks of the project will be scaled based on the individual's work rate and contribution to the project, as described in more detail in Section 5.2.

The module runs from the end of September until the middle of May - with Christmas, Easter and assessment breaks in between (term times are listed here - https://www.nottingham.ac.uk/about/keydates/index.aspx).

1.6 Supporting Lectures

The following lectures / sessions will help you through the process:

- Autumn Term:
  - Week 1:
    - Introduction to group projects (group project coordinator).
    - Team introductions (academic skills team).
    - Team dynamics (industry speaker - TBC).
  - Week 2:
    - How to write a CV (careers and employability team).
    - GRP background information (group project coordinator).
  - Week 3: EoI and Pitch preparation (group project coordinator).
  - Week 4: CV Feedback sessions (time and date TBC)
  - Week 5: Pitch week (20-10-2020, 14:00 – 17:30)
  - Week 7: Industry talk on project management (TBC)
  - Week 8: Industry talk, version control (TBC)
  - Week 9: Writing your interim report.
- Spring Term:
  - Week 1: Semester 2, introduction (group project coordinator)
  - Week 8: Writing reports
  - Week 9: Preparing for demo day
  - Week 11: Group project open day.

Times and locations for these lectures will be announced on Moodle and in your timetable.

1.7 Teams

Teams consist of 5 to 7 students and are composed such that members have different skillsets (e.g. highly technical, project management, interpersonal) and backgrounds. The diversity of the team will not only benefit the project, but it will also help team members to develop their mentoring skills, e.g. by sharing knowledge and experience in areas they are particularly strong in with other members of the team. Every team member is actively encouraged to do so and doing so is part of the goals for this module.

Note that you are not allowed to form your own groups, and that you are not allowed to swap places with someone from another group.

1.8 Roles and Responsibilities

1.8.1 The Group Project Coordinator

The group project coordinator is a member of academic staff who is responsible for the overall administration and coordination of group projects. Currently this is Dr. Geert De Maere. The group project coordinator will organise the project briefings, pitches, project allocations, support and deliver lectures, and maintain the project handbook. (S)he will also administer / coordinate the efforts of supervisors with respect to the sourcing and assessment of projects. Any queries concerning your individual project should always be directed to your supervisor(s) in the first instance. The group project coordinator will mediate if there are issues that cannot be resolved.

1.8.2 Supervisor Roles

Academic Supervisor

Every team is assigned an academic supervisor with whom they should meet up regularly throughout the year. Your academic supervisor is responsible for providing appropriate guidance to the team and should be kept informed of progress. However, teams are meant to be self organising. That is, the academic supervisor is not "in charge" of the project, nor there to make decisions on behalf of the team.

As a guideline, the academic supervisor should:

- Host meetings once every fortnight with demos, e.g. after every sprint. This will also encourage the team to follow an agile process with two week sprints.
- The meetings can focus on progress since the last meeting, problems that have arisen, and a plan for the work that will be carried out by the next meeting.
- If working on an industry project:
    - The academic supervisor should be copied on all important communication with the industry sponsor of your project.
    - The academic supervisor will liaise with the industry sponsor on a regular, e.g. monthly, basis. This could be, e.g., by joining once a month to the project meetings with the sponsor.
    - The academic supervisor will request informal feedback from the industry sponsor to inform the marking process.

The academic supervisor will be the main marker of your project. (S)he will assess and provide feedback for:

- The **bidding process**: the CVs, expressions of interest and the pitches. If working on an industry project, the assessment of the expressions of interest and pitches will be done jointly with the industry sponsor. However, it remains the responsibility of the academic supervisor to complete the feedback, submit it, and to set the mark.
- The **ethics forms**.
- The **interim report** and **prototype** that the team submits.
- The **group reports** (main report, software manual and user manual) and **individual reports** (main individual report and peer assessments).
- The **software** and the **management of the project**.
- The **stand / demo** on the group project open day, and the **demonstration video**.

The academic supervisor will also be given the opportunity to **moderate peer review scores** (given in the peer review process – see Section 4.2) to ensure that they are a fair reflection of the contributions of individual team members. It is therefore in the team's best interested to meet regularly with the academic supervisor.

Industry Sponsor

Industry sponsors offer support to the team between the middle of October and the end of May. The amount of contact can vary but, as a guideline, you should expect to **meet your industry sponsor for about an hour every two weeks**, in addition to pitch day and showcase day. These meetings can take place at the sponsor's premises, on campus, or remotely - e.g. using Skype or MS Teams. A combination of these often works well.

The role of the industry sponsor can differ. Some will primarily act as the client or product owner. Others get more involved, e.g. by taking up a mentoring role or by providing detailed technical input.

All communication with your industry sponsor should be in a professional manner and the academic supervisor should be copied in. Note that the industry sponsor will be asked for their input when your academic supervisor marks the project.

Room bookings

If working on an industry project, and when meeting with industry supervisors on campus, teams can book meeting rooms. This can be done by contacting the staff at the reception desk in the school of Computer Science (Gemma Singleton – gemma.singleton@nottingham.ac.uk, and Lindsay Norman – lindsay.norman@nottingham.ac.uk). Please state explicitly that the booking is required to meet with your industry sponsor, and specify the date, time, and equipment that you require. Note that all other (regular) meetings should take place in the labs or in other publicly accessible rooms.

1.8.3 Team Roles

Team Leader

Every team elects a team leader at their first meeting, or soon after. The team leader need not be the most technically skilled person in the group, but is probably someone who is organised, and has good people and communication skills. The ability to effectively organise, motivate, and listen to others will be far more important for this role.

The responsibilities of the team leader include:

- Project management:
  - Setting up, maintaining, and refining the backlog of user stories in conjunction with the product owner. The backlog should be accessible to everyone involved in the project, including your academic supervisor and industry sponsor.
  - Agreeing on the "definition of done" (and acceptance tests) with the product owner.
  - Preparing, guiding and time managing the sprint planning meetings and retrospectives.
  - Ensuring that work is divided fairly between team members.
  - Ensuring that the team has all the necessary information to complete the next sprint.
  - Monitoring progress of the sprint and keeping track on velocity and burndown.
- Leading communication with the sponsor and supervisors of the project.

Note that the Team Leader is also expected to contribute to other (technical) aspects of the project.

Group Administrator

Every team elects a Group Administrator (not the same person as the team leader) at their first meeting, or soon after. The Administrator makes sure all your work is there at the right time and it should be someone who can be relied on.

Responsibilities of the Administrator include:

- Making bookings and submitting reports and deliverables on behalf of the team.

- Taking meeting minutes, monitoring attendance of team members of meetings, and completing attendance forms in Moodle.
- Ensure that reports are proofread before submission.

The Group Administrator is also expected to contribute other aspects of the project.

Git Lead

Every team elects a Git Lead at their first meeting, or soon after. This person will be responsible for making sure the state of the Git repository is kept up to date and that no one commits to the main branch without passing all unit and regression tests. The Git Lead may be the same person as the group administrator. The best person is probably someone who has prior experience with Git, or who is prepared to get to know about it.

A lecture will be given on the use of Git. An excellent resource on the ins and outs of Git is available here: https://git-scm.com/book/en/v2.

The Git Lead is also expected to contribute other aspects of the project.

1.9 Expenses

1.9.1 Travel

Expenses to travel to the industry sponsor of your project will be reimbursed by the university. However, you are expected to use common sense and be responsible in this. That is, travel as much as possible (and practical) with public transport, or if going by taxi, share a taxi whenever possible. If traveling by car, please ensure that your insurance covers you for “work / business” related trips.

Approval must be obtained from the project coordinator (Dr. Geert De Maere) prior to booking any travel. Make sure you keep any receipts to claim back expenses. The expense reclaim form can be downloaded from the COMP2002 Moodle page.

1.9.2 Equipment

There is some budget available to purchase devices or services necessary for the project. The purchasing process involves three steps:

- The Team Leader (or Administrator) contacts the group project coordinator (Dr. Geert De Maere) with a list of equipment or services that are required, and a breakdown of the cost.
- The group project coordinator fills out a requisition for the products.
- School staff orders the products.

Approval must be obtained for all equipment prior to purchasing it and all purchases must follow the above protocol.

1.10 Background Reading

The following resources may be relevant in the context of your group project:

- *"Software Engineering"*, Tenth Edition, Ian Sommerville, Pearson, ISBN-13: 9780133943030.
- *"Pro Git"*, Scott Chacon and Ben Straub, APRESS.
- *Agile project management:*
  - *"Scrum: The Art of Doing Twice the Work in Half the Time"*, Jeff Sutherland
  - *"Agile Handbook"*, Emerson Taymor (free e-book)

2 The Project

2.1 The Bidding Process

2.1.1 Project Sign-Up

All project briefs are listed on the Moodle page for COMP2002. The project briefs give an initial specification of the project and may initially be vague. Teams are expected to shortlist projects that they want bid for as soon as possible (prior to the start of the sign-up process, which takes place late September to early October).

During the sign-up process, teams register to bid for projects on a FCFS basis. That is, if other teams register for a project before you, the other teams will be given the opportunity to bid for the project and your team will have to register for an alternative project. It is therefore recommended that teams shortlist about 6 projects in case they miss out on their first few options. Note that teams are responsible for jointly shortlisting the projects, but that only the group administrator will be able to register the team's interest for projects.

To maximise the chances of teams getting at least one of their preferred choices, the sign-up process is spread across three consecutive days. That is, you register to bid for the first project on day 1, the second project on day 2, and the third project on day 3. The different registration systems will open automatically at 09:00 on each day (the exact dates are announced in Moodle). When the registration process closes on the third day, every project will have three teams against it, and every team will be signed up to bid for three projects.

2.1.2 Bidding for a Project

A bid for a project consists of three parts:

- A CV for every member of the team.
- An expression of interest (EoI) for the project - using the template provided in Moodle.
- A pitch to the sponsor and supervisor of the project.

Every team is expected to submit three different bids, one for each of the projects they have signed up for. Note that the EoIs and pitches should be tailored to the different projects, but that the same CVs can be submitted for the different bids.

Curriculum Vitae

Every team member is requested to submit a Curriculum Vitae (CV). This CV should be tailored to the role you apply for (in this case, software engineer) and provide convincing evidence that you should be considered for the project. A lecture by a member of staff from the Careers and Employability Office will provide guidance on how to write a CV.

CVs will be marked by an academic, and the mark for your CV will count towards the individually assessed components for the module (see Section 5). In addition, your supervisor will provide feedback on the strengths of your CV, and the areas where you could make further improvements. This feedback will hopefully be helpful when you decide to apply for internships or for jobs in the future.

If you are bidding for a project with industry, and if you give us permission to do so (in Moodle), your CV will be shared with the industry sponsor of the projects that your team applies for. You are not obliged to share your CV with industry sponsors and opting out will not affect your marks.

An example of a good CV can be found in the appendices.

Expression of Interest

A good Expression of Interest (EoI) should:

- Motivate why the team is bidding for the project and wants to work with the sponsor.
- Demonstrate some level of understanding of the problem and its context.
- Demonstrate technical understanding and illustrate that the team has the necessary skills to carry out the project.
- Give an understanding of how the project will be managed.
- Be tailored to the project and honest on what the team can offer.
- Discuss Legal, Social and Ethical aspects of the project.

The team leader is responsible for fairly allocating the workload to write the EoIs to the different team members, whilst giving every member the chance to provide input and review the different EoIs in the process. The team administrator is responsible for submitting the EoIs.

A lecture will provide additional information on how to write EoIs.

The Pitch

Every team will pitch for 3 projects, and 3 pitches will happen for every project. Your pitch should convince the project sponsor that your team is “the best team” for their project. A good pitch should:

- Contain short introductions of the team members, their roles within the team, and how the project will benefit from members’ backgrounds (cover both technical and soft skills).
- Have all members of the team involved in the pitch acting as a united team.
- Have relevant content, mockup models, a workplan, work packages, and proposed milestones.
- Explain project management approach that will be used.
- Be tailored to the brief, well structured, supported by slides, visuals, and animations.
- Be engaging, interactive, professional, and enthusiastic, whilst demonstrating ambition and realism.
- Give strong and concise answers to questions.
- Contain conclusions and a final summary.

Handouts are often appreciated and the use of company logos without permission must be avoided.

The project sponsor and / or supervisor will evaluate all pitches and EoIs for their project and list their order of preference for the teams. This order will be an important consideration when allocating teams to projects.

2.1.3 Award Process

The preference order based on the CVs, EoIs, and pitches is an important factor in the allocation of teams to projects. If a team is ranked first for multiple projects, the team’s preference for projects will be taken into account. The allocation of teams to projects normally happens within two days of pitch day.

2.2 Project Management

A major component of the project will be to practice software engineering techniques and technical skills that you have covered theoretically or at a lower level of complexity in other modules. This includes programming skills, software design, team skills, and project management

An Agile project management approach, which is in its purest form typically well suited for smaller projects with co-located teams that communicate informally, is recommended. Agile methods have a strong focus on delivering a finished piece of work (e.g. a working product or completed analysis) after every sprint (to be demo-ed to supervisors during meetings) and typically interleave requirements elicitation with development, rather than having a heavy weight upfront requirements analysis (as with plan driven approaches) that causes significant overhead. In summary, agile methods encourage customer involvement, embrace changes to requirements, delivers small and frequent releases, focusses on simplicity of software and processes, and assumes self-organising teams.

There must be evidence of the project management process you have used. Deliverables should be planned on a two week cycle through term time and results of demo days and retrospectives (see below) should be recorded. Note that the project management process used is part of the assessment criteria.

2.2.1 “Backlogs”

A formal “to do list” or “*product backlog”* is maintained during the project. This list contains a set of “business objectives” in the form of user stories, product requirements, engineering improvements, and supplementary tasks, worked out with the sponsor/supervisor at meetings. Items in the backlog are prioritized by the Scrum Lead (team leader) in conjunction with the customer (project sponsor). A subset of the highest priority items in the backlog is selected by the team to form the *“Sprint backlog”* containing the set of objectives to be addressed in the next development iteration (see below).

Note that:

- Microsoft Planner, Trello, Jira, just to name a few, help to maintain backlogs.
- Your supervisors should be given access to the backlogs.

2.2.2 “Sprints”

Two week development iterations or “Sprints” are recommended. Items are selected from the product backlog to plan the “Sprint” during the planning meeting, together with objectives for what will be demonstrated at the end of that sprint. Tasks are checked out by team members with responsibility and recorded as completed by moving them to a done list.

The *“burn down”* is calculated to keep track of progress. At the end of the Sprint, a demo will take place. A *“retrospective”* or review meeting is held after the demo to discuss:

- what has gone well?
- what could be done better?
- what is holding the team back?

Retrospectives should be held in a constructive manner, giving constructive feedback on how to improve the working of the team.

Note:

Please ensure that targets for next sprint become gradually more realistic (e.g. time scales) and inform your supervisor/sponsor when other class activities (coursework for other modules, for example), exams, and holidays are planned.

2.2.3 “Stand-Ups” or “Scrums”

Informal team meetings (known as “stand-ups” or “Scrums”) should take place on a regular basis and should not take longer than 15 minutes. Whilst it is recommended that they take place daily, this may not always be practical in the context of group project. Stand-ups review progress and prioritise work. They are commonly guided by three questions:

- What have you completed since the last meeting?
- What do you plan to complete by the next meeting?
- What is getting in the team’s way?

2.2.4 Supervisor Meetings

Teams are expected to meet regularly with their academic supervisor and industry sponsor - if applicable (see Section “Academic Supervisor” and “Industry Sponsor”). Meetings are organized by the team lead or group administrator, who ensures that all communication happens in a professional manner.

Note that:

- Attendance at formal meetings is compulsory - non-attendance may affect your individual marks.
- Meetings should be documented with attendance records and meeting minutes (by the group administrator), made accessible all team members and supervisors (e.g. stored in sharepoint, google drive, or similar).
- The main computer science lab (A32) is booked for 2 hours every week (see timetable for the exact time), and no other activities are scheduled during this time (all team members should therefore be able to attend).

2.3 Version Control - Git

Teams are required to use the school's GitLab server (https://projects.cs.nott.ac.uk). If the sponsor requires the use of a different version control system, the repository should be mirrored regularly onto the school's GitLab server. A workshop on Git will be given by an industry partner and additional do on the use of Git can be found here: https://git-scm.com/documentation.

All teams-members have a GitLab account and a repository has been pre-initialised for every team (https://projects.cs.nott.ac.uk/COMP2002/2019-2020/team<teamnumber>_project.git). The Git repository is managed by the Git Lead (see Section "Git "). Regular commits of developments should be made by team members, with evidence of merging when appropriate. Note that any other documents, e.g. meeting minutes, attendance sheets, etc., should not be stored in GitLab, but in Google Drive, Sharepoint, or OneDrive, for instance (and not on Facebook).

If you struggle with accessing Git or Multi-Factor Authentication (MFA), please contact the IT Smart Bar. Resetting the Azure MFA account frequently resolves the problem.

External partners can be added to the Git repositories. However, they will have to apply for an associate's account with the university first. Further information is available here and here (internal only). If and external partner for your project requires access, please contact the group project coordinator to start the process.

2.4 Code quality

Code quality takes into account, e.g., whether:

- Is the software well commented?
- Does it have a good and clear layout?
- Have you used consistent and appropriate naming conventions?
- Can someone else easily understand your project software so they could extend it?
- If, for example, you are using Java, have you used Javadoc compatible comments?

2.5 Testing

Software should be developed using a test-driven development approach where appropriate (some user interface components may be difficult to unit test so you may need to have clear user test patterns). Where possible – in particular where a UI is being designed – beta testing should be reported. Bug reports should always be listed. The structure of the repository should reflect testing, e.g. by using a delivery branch, development branches and issues. Evidence of all software testing should be recorded.

2.6 Open Day

Open day takes place in the Summer term, usually in the first half of May. All teams demo their work in a "trade fair style event", taking place in A32. This involve preparing an (interactive) demonstration of your software, a poster, and a nice display/stand.

The open day will help you to develop your communication skills and is attended by numerous industry visitors, academics, and other CS students. It is an opportunity to show off the results of your efforts and hard work during the year. It is also a perfect opportunity to make connections with industry partners for future graduate opportunities, internships, etc. Open day is often seen as the highlight of the year.

Prizes (goodies and/or financial rewards) will be given out by industry sponsors for *“the overall project“*, *“the runner up”*, *“the best demo”* during demo day. Prizes are independent of the marks you get from the staff markers.

2.7 End-Of-Contract

The end of contract required the team to write a group report (including software and user manual). In addition, every team member will write an individual report and complete a peer assessment. These deliverables are described in more detail in the relevant section below.

2.8 Golden Rules

Below are some tips based upon the reflective comments in the final reports from previous years.

- Work consistently throughout the project. Start work on your project quickly, and don't leave all the real work until just before the deadlines. The agile process with bi-weekly sprints aiming to deliver a demonstratable result at the end of every sprint is designed to encourage you to do so.
- Make sure that you have enough to do. Don't wait to be assigned tasks - volunteer for things that you find interesting. Keep in mind that the peer assessment is used to scale group marks, and that underperformance of individual group members can significantly influence their marks. That is, take on an active role in the project and work hard.
- Use an agile process in which design is embedded with implementation during the course of the process. Do not leave the programming until too late.
- Use test driven approaches and do not leave integration testing until the end.
- Use a SCRUM board (or similar) to keep track of progress. This will help to make sure that everyone in your group is working consistently. Your supervisor can provide some guidance with managing your project. Ultimately, however, it is the responsibility of the group themselves to monitor the progress of each member, to ensure that tasks are being carried out as required, and to take remedial action as and when appropriate.
- Take sprint meetings and retros seriously and use efficient communication (as explained in the team workshop).
- Keep good records of sprint meetings and retros and share them with the rest of the team and supervisors. This will help to keep meetings efficient and will prevent going around in circles.

3 Group Reports

Group reports are produced by the team. The team's administrator is responsible for submitting one copy on behalf of the team to the relevant submission system on Moodle.

3.1 Submission Format

Reports should follow a pre-defined "format" for the front page, of which a copy is available for download on Moodle. The style itself can be changed to match the rest of the report. The front page of your report must contain:

- Project information:
  - The project title.
  - Project sponsor (e.g. UoN or industry partner).
- Team information:
  - The team number.
  - The team members (first name, last name, student id, username).
- Supervisor information:
  - Academic supervisor (first name, last name).
  - Sponsor company.
  - Industry supervisor(s) (first name, last name).

The second page of the report must contain the documentation links to the:

- Trello board.
- Code repository.
- Document repository.
- Further online documentation (if applicable).

Note that permissions to access the above resources must be granted to the team supervisor.

All reports must be submitted in *PDF Format* to the appropriate submission system *on Moodle*. The file name must be "*teamX-groupReport.pdf*" (replacing the X with your team number).

3.2 General Advice

When writing reports, please take the following general advice into account:

3.2.1 Writing Process

- Ensure that there is one editor in charge of the documents, preferably someone with an emphasis on presentation and good English. This helps to write reports in a uniform style, even if individual sections are drafted by different members of the team.
- It is recommended to avoid cutting and pasting sections written by other team members, and to instead read and rewrite them. This will help to improve coherency across the reports.
- Ensure that the reports are ready well in advance. It is reasonable to expect that you will require several writing and editing iterations to obtain good reports, including after feedback from your supervisor.

3.2.2 Content

- Take into account that this is a software engineering group project. That is, in addition to the deliverables and technical achievements, your reports and supporting documentation should demonstrate the use of established software engineering practices and good project management.
- The reports should focus on key aspects, insights and observations. Avoid introducing common knowledge (e.g. no need to explain class hierarchies in Java) or stating the obvious.
- The reports should not be written as a rapport of the chronological sequence of events carried out by the different members of the team during the project.
- If explaining bespoke algorithms developed as part of the project, please introduce them as pseudo code and focus on their key functionality.

3.2.3 Language & Style

- Write your reports from the perspective that you are working for a software house that is delivering a project for a high-profile customer. Consider what impression your reports / manuals would create and write them with the intention to maximise the chance of repeat business with the client.
- Ensure that language is formal and professional. Avoid the use of emotive and subjective statements.
- Additional guidance on technical writing is provided on Moodle, including a guide on grammar / spelling / style – put together by Prof. Brailsford. Please consider the advice given in this document carefully.

3.2.4 Structure

- Please make sure that a table of contents and an outline section is included. The outline section should be kept short and accessible to the lay person.
- Good structure, efficient use of diagrams and illustrations to support written text, and good referencing are the cornerstones of good reports.
- ***By no means should the categories or content items listed in the sections below be interpreted as the section headers that you must adhere to.*** In contrast, students are expected to carefully consider the narrative and flow of their reports, which makes it likely that you will have to deviate from the apparent structure listed below.

3.2.5 Referencing

- Ensure appropriate and consistent in text referencing to technical resources and the scientific literature. Use a consistent referencing format, both in the text and for the full references at the end of your reports.

3.3 Legal, Societal, Ethical and Professional Issues (LSEPI)

3.3.1 Context

Your project doesn't happen in a vacuum. Your work is affected by the world around you. This influence includes laws and University policies, but also the concerns and priorities of society and your peers. On the other hand, your work can affect the world around you. In fact, it might also change the world (for better or worse)!

As part of your "Contributions and reflections" in your interim and final report, explain how the team has considered the "bigger picture" within which your work is situated. The checklist, below, identifies several issues. Depending on the project, some of them may be more relevant than others. But everyone should have something to reflect on (even if it is the last question).

In the interim report you should highlight which issues are the most significant for your project, and how you will address them in the remainder of your project. In the final report you should present your reflections on each issue, including why they are (or are not) significant for your project.

3.3.2 "Big picture" Reflection Checklist

Intellectual property (the Copyright, Designs and Patents Act (1988))

- Has the work created any intellectual property? E.g. source code, documents, photos or recordings, patentable technologies, trademarks. If so:
  - What intellectual property and who owns it?
  - Do you have a plan to commercially exploit the work? How?
  - Do you plan to distribute or promote the work in other ways, for example using open source/open data/creative commons licenses? How?
- See also "Intellectual Property Rights" in the 3rd/4th year project handbook and "Provision and Processing of Intellectual Property Rights for Students and Graduates at The University of Nottingham"

Research ethics

- Does the work involve human participants or data subjects? If so:
  - Have you followed the University of Nottingham research ethics requirements?
  - How does the work safeguard participants and data subjects?
  - How does the work reflect principles of informed consent and voluntary participation?
  - Were there any ethically tricky aspects to the work, and how did you deal with them?
- See also project module guidance on research ethics and https://workspace.nottingham.ac.uk/display/CompSci/Research+Ethics+Guidelines+for+Academic+Staff%2C+Researchers+and+Students

Data protection (the Data Protection Act (2018))

- Is the work collecting or using personal data?
- If so, how is it complying with the requirements of data protection legislation?
- Note, this should also have been covered by your research ethics

Other legislation:

- Is the work affected by other legislation? If so, how is the legislation relevant and how has the work responded to that legislation? For example:
  - the Equality Act (2010) has implications for accessibility,
  - the Computer Misuse Act (1990) and Counter-Terrorism and Security Act (2015) have implication for (potentially) criminal use,

  - the Online Safety Bill (if/when it comes into force) will have particular implications for systems accessed by children and other vulnerable groups,
  - the Privacy and Electronic Communications (EC Directive) Regulations (2003) have implications (together with the Data Protection Act) for communication systems.

Broader ethical and social considerations

- How have you considered the broader ethical and social aspects of the work? For example:
  - Who will benefit from the work? How?
  - Could the work have unintended consequences? What?
  - Could the work have military or criminal applications? What?
  - Who might be affected by the work, directly or indirectly? How?
  - Have people who might be affected been able to influence the work?
  - Is the work accessible or beneficial to all?
  - Is the work likely to reduce or increase societal inequalities?
  - What impact might the work have on the natural environment?
  - Is the work worth doing? Why?
- See Responsible (Research and) Innovation (COMP3020). The RRI Prompts and Practice cards, (https://experience.horizon.ac.uk/2022/09/30/rri-prompt-practice-cards/) and the Moral-IT cards (https://lachlansresearch.com/the-moral-it-legal-it-decks/) may also help you to reflect on these aspects of your project.

3.4 The Interim Report

The interim report is a group report that is due around the half-way stage of the project. It should contain 3000 - 4000 words (around 6-8 pages, excluding any appendices). The main content should reflect (where possible) the assessment criteria below. Note that, as stated above, it is not the goal that you stick rigorously to the headings below, but that you ensure that the narrative of the report flows well.

- Project Background & Understanding (10%):
  - Evidence of understanding of the initial brief / specification
  - Evidence of own interpretations / additions / innovations
  - Depth and context of the problem presented (e.g. existing products)
- Requirements & Critical Analysis (15%):
  - Depth / clarity / coverage and appropriate prioritisation of requirements.
  - Motivation / reflection for changes to requirements and scope reductions.
  - Results of focus groups / user surveys (where applicable)
  - Analysis of hardware / tools / methods / software libraries for the project and motivation for the choices made.
- Initial software implementation & testing (30%) – where applicable
  - Architecture / design / proof of concept implementation
  - Data storage / management / manipulation / use of (personal) data (if applicable)
  - Appropriate use of version control system (branching / merging)
  - Evidence of testing
  - Preliminary inline software documentation
- Project Management & progress (15%)
  - Evidence of effective / appropriate project management (Agile, SCRUM, SCRUMBAN, Kanban, etc.)
  - Documentation of sprints / retrospectives / workload management / formal meetings (where applicable)
  - Use of project management tools (Trello / Microsoft Planner / Teams / Slack)
  - Effective use of team skills / skills transfers / cross training
  - Professional attitude / disciplined / consistent / timely / punctual / autonomous & self organising
- Preliminary Reflection (15%)
  - Tools / techniques / methods / technical difficulties / testing
  - On project management / current progress / encountered problems / mitigation measures
  - LSEPI (see Section 3.3)
  - Functioning of the team so far and contributions made by individual members
  - Forward plan that addresses concerns identified
- Style (8%)
  - Accessibility / clarity & motivation of the argument / structure (e.g. by key parts / components of the project)
  - Concise / conceptual descriptions / appropriately balanced content / length / depth
  - Grammar / spelling / articulation / formatting / structure / jargon & definitions
  - Use of visuals / graphics
- Resource links (2%)
  - Links to document repository / project management documentation (e.g. Trello / Microsoft Planner) / code repository
- Conclusion & Summary (5%)

- summary of key points and future plan

3.5 The Final Report

The final report consists of three key parts:

- The main report.
- The software manual.
- The user manual.

All three parts must be submitted as one single document. Additional guidance on how to write each of the individual parts is given below.

3.5.1 Main Report

The main report, the first part of your final report, builds upon the previously written interim report, from which content can be re-used to present a holistic document describing the project and its results. It can be reasonably expected to contain between 6000 and 8000 words (around 12-16 pages, excluding appendices). It documents - in addition to the interim report's improved and updated content - the progress made during the second term.

The main report provides general project background (similar to what was included in the interim report). Whilst some sections of the interim report may remain relevant, key changes to sections from the interim report (e.g. to reflect recent progress), newly added sections, or any sections that have been removed should be listed in a "change section" at the start of the main report in line with common practice.

In addition to the above, the main report should have an increased focus on constructive reflection and software engineering. That is, any technical content should, where appropriate, be supported and documented using established methods from the domain of software engineering, for instance when introducing user-, functional- and non-functional requirements. In relation to the reflection element, the following items should be considered (where applicable):

- A reflection on the success of the project and the achievements of the project by the end of contract.
- A description of how requirements have changed over the course of the project and how the team dealt with this.
- A description of what changed in the methods used and the team's approach (even if the requirements did not change).
- A motivation behind the technical approaches that the team used in the implementation and a motivation for them.
- A reflection on what went well, what was hard to achieve, what went badly, what the team has learned from this, and how the team's functioning could be improved.
- An overview of how the work was split between different group members.
- A reflection on the management of the project (sprints, sprint planning, retros).
- A reflection on how contingency measures helped to deal with unexpected circumstances, or what contingency measures you would take in hindsight to better deal with unexpected circumstances.
- A reflection on LSEPI (see Section 3.3) elements identified in the interim report.
- Promising future directions for the project.

The above list of items is not exhaustive and teams are expected to reflect on additional aspects. That is, they should use their own judgement on what other content to include, or which items to not include from the list above (if they are not relevant in the context of your project).

Finally, any feedback received from your team supervisor previously on the interim report should be carefully considered and addressed.

3.5.2 Software Manual

The software manual is intended to help future developers maintain and extend the existing code base. There is no restriction on the length of the software manual, however, in most cases it would not exceed 20 pages.

The content of the software manual will be highly dependent on the project. It is the team's responsibility to consider what to include (some suggestions are listed below). In all cases, any content included in the manual should be documented using established methods from the field of software engineering where appropriate. Examples include, use cases, sequence diagrams, activity diagrams, class diagrams, data flow diagrams, entity relationship diagrams, etc.

The focus of the software manual should be on the conceptual implementation of your code. It should be presented at the level of an experienced programmer. Detailed information on specific code should be included in the inline / online documentation of your code rather than the software manual. The exception to this is if you have developed complex bespoke algorithms as part of the project, in which case the functionality of the algorithms should be explained in a conceptual manner using pseudo-code (rather than programming code).

Examples of other items to consider and include in the software manual, where appropriate, are:

- A brief introduction to the problem, its goals, and its scope.
- Definitions of project specific terminology and jargon.
- An overview of the high level structure / concepts that underpin the code.
- The architectural design of your implementation and a high level logical / physical architecture.
- An overview of data storage mechanisms / technologies / frameworks that were used, external dependences, libraries, proprietary systems and project specific hardware / software (with references to relevant documentation).
- Coding conventions, maintainability conventions, and design patterns that you have used.
- A description of project specific testing approaches and frameworks that you have implemented and an introduction on how to use them.
- A programmer installation / build manual detailing how to install your code.
- Links to further online documentation, if available.

Teams working with an industry partner are advised to consider the sponsor's suggestions on the software manual.

3.5.3 User Manual

The user manual is meant to describe the use of the software product that you have developed at a level that the expected user / audience would be able to understand. Its focus is on how to use the product that you have developed, as opposed to how it is implemented (which is documented in the software manual).

The audience can differ from project to project. In some cases this will be, e.g., system administrators, in other cases this will be, e.g., gamers. If the users are expected to be non-technical, the manual should be presented in a non-technical manner, understandable to the lay person. If you have developed a product meant to be used by experts in the field, the user manual could be presented in a more technical manner.

A good user manual is well structured, comprehensive, and relevant. Items that could be included in the manual depend on the type of the project. Examples include: an introduction to the goals of the product; an installation and configuration manual; how to get started; an overview of hardware and software requirements; a comprehensive description of the product's functionality; a troubleshooting guide; data gathering and data protection policies; frequently asked questions; caveats; etc. Note that this list is not comprehensive, and that additional items may need to be considered for some projects.

In terms of its style and presentation, the manual should be pitched at the right level, be easy to understand, appropriate in length, and well structured. The content should be balanced and supported by graphics where appropriate. References to online video tutorials or other media content produced by the team may also be included.

Please use a clear and professional layout appropriate to the type of product that you have developed. For instance, in the case of a game, one would expect the user manual to have highly creative content. In the case of a, e.g., tool for system administration, the content will be more formal and business oriented.

3.5.4 Demonstration Video

A 5-minute demonstration video (which is also assessed) should be submitted as part of the final report. The demonstration video must be encoded and submitted as MP4. This is a standard format that can be generated from most video software on all platforms. Videos submitted in other formats may be rejected.

The focus of your demo video could include the key goals of the project, its context, the requirements, the implementation, the realisation, the progress by the end of contract, and potential future directions. The video should be used as an opportunity to "sell" (but not oversell) the work that you have done and product that you have developed. It should highlight the key achievements of the team.

As a general guide for the demo videos:

- Display the team number and the different team members on the first video frame, together with the title of your project, supervisors, and sponsor.
- Prepare the narrative for the demo video before the recording.
- You can edit pieces of narrative to produce the final demo video.
- Speak clearly and be creative.
- Use audio-visual aids as appropriate illustrating crucial features of your work.

3.5.5 Abstract

Teams are asked to provide a short abstract (5 – 10 lines) that will be included in the programme distributed for the showcase. The abstract should document what the project was about, why it matters, how it was approached, and how it can be built upon in the future. A template for the abstract is provided on Moodle.

4 Individual Reports

Individual reports are written by the students individually. That is, every member of the team writes their own reports and submits them to the relevant submission systems on Moodle.

4.1 Individual Report - Reflective Statement

Your individual report focusses on your work within the group and the challenges that you may have faced. Its main purpose of is to reflect on the quality of your own contribution, the experience you had in the team, and lessons learned from it for the future. Your report should focus on both the technical contributions that you have made as well as the interpersonal role that you have played. Please contextualise your achievements against the requirements for the project.

A template for your individual report (that must be followed) is available on Moodle. The template contains five sections:

- *Why the group applied for the project*, focusing on how the interest, technical skills, past experience, and future aspirations influenced the choice for this project. This should reflect the team's view as well as your own personal perspective.
- *The accuracy of the original project description*, giving you the opportunity to express how, in your own view, the original project description and requirements have evolved during the course of the project. This gives you an opportunity to demonstrate your own understanding of the project.
- *The role that you have played within the team,* focusing in a concise manner on the technical, administrative, managerial, and interpersonal contributions that you have made to the project and team. Your focus should be on your key contributions and, where possible, be supported by contextual "evidence". How did you use your strengths to the benefit of the team and project? What were the different roles that you took on within the team? Were you primarily technical, managerial, or both? Were you a key player in the successful completion of the project and the operation of the team, a regular contributor? Was your work rate consistent? What initiatives did you take, how was your time management, how did you collaborate with other members to help them to achieve their best?
- *Difficulties that you experienced within your role,* focusing the technical challenges that you encountered as well as in any other aspects of your role.
- *Difficulties that you may have experienced working in this group and constructive actions taken to resolve these challenges*, documenting any challenges that were encountered by yourself or the whole team, and the constructive / proactive measures that you took to address concerns to the benefit of the team. Are there any things that you would do different in hindsight, what lessons have you learned for the future, how would you expect your current experience to influence your future work.

Please try to keep your answers concise, to the point, factual, and make sure that they cover all *key aspects and contributions* you made to the project in your role. Be honest, accurate, and realistic in your reflection and demonstrate that you have self-awareness and a good understanding of the functioning of the team. Finally, motivate your answers well, avoid too much focus on minor details, and avoid the use of emotive statements.

4.2 Peer Assessment

Peer assessment is an opportunity to reflect (in a constructive and objective manner) on the work of other members in the team. You will give each member a peer review score which, after moderation by the supervisor, will be used to adjust the group marks (see Section 5.2) per team member. A written justification supporting the peer review scores is also required for each member. ***The peer assessments are submitted in complete confidence, i.e., they are not shared with other team members.***

A peer review template (that must be followed) is provided on Moodle for each student. The following aspects are considered when judging your fellow students' contributions:

- *General Contribution* to the project: coding, relationship with "customers," data gathering and so on.
- *Creativity*: introduction of new ideas for the project, contribution to designs etc.
- *Work rate*: did they do the work given to them and complete their fair share, or did others have to substitute for them? Did they perform the necessary background research? Was work delivered on-time and of high quality? If they did not attend, they may still have delivered the required work as and when desired.
- *Team membership:* how did they perform as a team member? Did they communicate clearly with other team members, openly express opinions, listened to other views, share opinions and knowledge, and adopt other approaches when appropriate? Were they committed to the team goals, respectful to other members? Did they provide help to other members when necessary? Someone who did a lot of coding but did nothing to help the others to play their role should not get extra credit.
- *Attendance* at meetings & team work sessions.

It is not just the amount of contributed work that is being evaluated, say in terms of words in the reports or lines of source code, but a wide spectrum of contributions all of which are important for a successful end result. This means that everyone gets a chance to contribute according to their specific skills and get proper recognition for this. Clearly, someone who excels in designing the system architecture and implementing it has made a very important contribution. Likewise, someone who is a great writer and took the lead on getting the group reports together has also made a big contribution. But someone who was instrumental in mediating between conflicting views, and thereby helped holding the group together has also made a very valuable contribution to the group as a whole.

The contribution of each peer is rated on a 5-point scale ranging from 1 for "None" or "very poor" via 2 for "Inadequate", 3 "Adequate", 4 "Superior" to 5 "Excellent", where "Adequate" means that the assessed person has done what is expected: no less, no more. Note that it is the relative performance of group members that matters as the peer marking only serves to redistribute the assigned collective group mark. For example, if everyone rates everyone else "Excellent", the end result is that everyone gets the same mark for their contribution to the group work which is going to be equal to the collective group mark.

Assessing your peers is a privilege and big responsibility. Be fair, objective and constructive in your evaluations. Do not come to any agreements with your peers to "give each other 3 or 4" or any other thing. Make sure you make your judgements based on your real views. How well you express your opinion of your peers' contributions will have an impact on your own individual report. You also will not be able to control what others actually write and the marks you give. People are not always able to say what they really think if they are face to face with you.

Members of the School and Faculty have been very impressed with the quality and honesty of the peer assessment in the past, and there have been few problems. Each supervisor is charged with carrying out a sanity check on the peer marks based on what they have learnt during the interaction with the group members throughout the year, as well as from the group and individual reports. Should there be obvious problems with some of the peer marks, then the supervisor together with the module convenor will override any or all of those marks where necessary.

An example (fictitious) peer review can be downloaded from Moodle.

5 Assessment

The ***total module mark*** received for COMP2002 will be composed out of (weighted) ***marks for the team*** and ***individual marks***, as explained below. For projects with industry, the opinion of the industry sponsor will be taken into account by the academic supervisor when setting the marks.

5.1 Summary of Deliverables

The group project has no formal written examinations. Instead, it is assessed through a number of different deliverables that are completed at specific times during the project. These deliverables include:

- The bidding process:
  - Your CV.
  - The team's Expressions of Interest.
  - The team's pitches for projects.
- Ethics form(s).
- The project:
  - The code repository on GitLab (https://projects.cs.nott.ac.uk).
  - All minutes and other non-software documents as well as regular commits of developments. Progress on development will be assessed at the end of semester one and again at the end of the module.
  - Code/software including testing and documentation (for example Javadoc).
  - Project management documentation – e.g. Trello board, meeting minutes.
- Reports:
  - The team's interim report and prototype.
  - The team's final report that documents the requirements analysis (and reflection), the software documentation, the user manual.
  - Your individual report and reflection on your work.
  - Your peer assessment statement of team members.
- The Open Day – formal open day in the form of a trade show.

5.2 Group Marks

The group mark represents 80% of your total module mark. The group mark takes the following components into account:

- The expressions of interest and pitch (15% of your group mark)
- The final report (25% of your group mark)
- The stand, presentation, and posters on demo day (15% of your group mark – no poster is required for the virtual open day)
- The software, the inline documentation, and *"the process"* used to get the result (45% of your group mark)

The group mark will be scaled to calculate using a weight factor based on peer assessment. This can increase or decrease the mark for individual students and will be moderated by the supervisor to ensure fairness. That is, the final mark for the module will take into account how much individual students have contributed to the "shared parts" of the project.

5.3 Individual marks

The individual marks represent 20% of the total module mark for individual students. The individual marks are based on the following components:

- Your CV that you have submitted (20% of your individual mark)

- The quality of your individual report (60% of your individual mark)
- The quality of the peer review (20% of your individual mark)

5.4 Total marks

The total mark for an individual student is based on the (scaled) group mark and his/her individual marks. The total module mark uses an 80/20 weighting for the group/individual marks. The contribution for each of the component marks to the total module mark is shown below.

	Group Mark Weighting (out of 100)	**Weighting for Total Module Mark**
EOI and Pitch	15%	12.00%
Ethics documentation	-	-
Interim Report	10%	8%
Final Report	25%	20.00%
Stand/Demo/Presentation	15%	12.00%
Software/Project Management/Decision Making	35%	28.00%

	Individual Mark Weighting	**Weighting for Total Module Mark**
CV	20%	4.00%
Individual Report	60%	12.00%
Peer Assessment	20%	4.00%

Important:

- Ethics documentation will carry no actual mark, however, non-completion of this documentation (and follow up documentation for projects considered to require it) will nullify any marks obtained for any other components. Not going through appropriate ethics clearance is considered as academic misconduct
- Late submissions, be it of printed copies or electronic copies of reports and software, will be penalised at the standard university guidelines of 5 % per working day.

5.5 Re-assessment

If you are required to take re-assessment for the module (either due to ECs or for progression), you will be asked to:

- Consider the code and documentation (report, manual, etc.) that have been submitted already, and the initial and adopted requirements.
- Add a useful fix or addition to the code
- Update the documentation appropriately
- Provide a brief (1-3 page) report summarising what you changed, why it was changed, what effect it had
- A reflection upon what you have done and (if you so wish, to enhance the potential for reflection) within the initial project duration.

Your additional work should demonstrate:

- ability to work with (i.e. adapt/improve) the code and documentation already provided by others (and you)
- knowledge of software engineering methodologies
- ability to understand the requirements and specifications (as originally given or adapted)
- ability to select and deploy tools and techniques
- the ability to reflect upon your own progress, strengths and weaknesses

The marking criteria for the regular assessment will be used as a guide to evaluate your work.

All code changes should be pushed to Git, and the report should be submitted to the relevant submission system on Moodle.

Note that the student is responsible for contacting the project supervisor as soon as possible to discuss project specific details of the re-assessment.